

Once Upon a Sea

A poetic, interactive XR documentary

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ABSTRACT

ONCE UPON A SEA is an immersive virtual reality experience that transports participants to the Dead Sea, and provides access to one of the wonders of the world that has become inaccessible in the past 35 years. The Dead Sea carries a rich history, an undeniable healing powers, and an indescribable magnetism that can now be experienced in a virtual format. The experience takes place in photoreal volumetric captures of some of the most significant and beautiful sites in the Dead Sea, highlighting its beauty, diverse inhabitants, and its recent demise. In the past 30 years, the Dead Sea has receded dramatically due to human intervention and political neglect. The sweet water that fed the Sea was used for irrigation and potash evaporation pools left behind a ravaged land, ridden with sinkholes. Today, all beaches but one are inaccessible to the public due to these dangerous sinkholes, representing Israel's worst ecological crises. The destruction is progressing quickly, causing many socio-political battles as well as financial & personal distress to local residents and individuals who have dedicated their lives to the Sea. Israelis, Jordanians, and Palestinians, all whose countries border this body of water, have been affected by the demise of the Dead Sea. If nothing will be done, the Dead Sea as we know it will be gone for good. ONCE UPON A SEA is our call to action.

CCS CONCEPTS

• **Applied Computing**; • **Art and Humanities**; • **Interactive learning environments**;

KEYWORDS

virtual reality, environmental, documentary, immersive

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1 INTRODUCTION

ONCE UPON A SEA is an immersive virtual reality experience that transports participants to the Dead Sea, and provides access to one of the wonders of the world that has become inaccessible in the

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Once Upon a Sea is a requiem for the Dead Sea, once one of the world's greatest wonders. It's a poetic, interactive documentary in Virtual Reality using photo-based 3D models that share the stories of the people surrounding the Sea. Through physical exploration of the sea's forbidden landscape, to curious meetings with local characters introducing the social and ecological complexities of the region, the user will experience a rare glimpse into one of the world's most dangerous, soon to be extinct wonders. Floating in the Dead Sea is an experience equivalent to our first moments in life, inside the womb. This experience exists as a faint mirroring replacement to the physical visit to the beach which can no longer be accessed. This is a linear experience that uses natural Interactivity for the purpose of documentary storytelling, guiding the viewer through a maze of stories and sites. For the first time, the Dead Sea, has been captured using Photogrammetry and 360 Filming. In ONCE UPON A SEA, extremely intricate Volumetric Captures and 360 Films become a private place of exploration and experience.

In the Jordan Rift Valley, situated at the mouth of the Jordan River lays the Dead Sea, known as the lowest and saltiest body of water in the world. Dated 1.64 million years ago, the sea forms part of the border between Jordan and Israel. For thousands of years the Dead Sea has attracted visitors from around the Mediterranean basin. It is a prospering tourist region accommodating close to a million visitors a year; but not for long.

For an exceptionally large part of human history, the Dead Sea has been considered a prime example of an environment completely devoid of life, a condition which during the Greco-Roman period was ascribed to natural causes—the bitterness of the water and the hydrogen sulphide stench associated with the area. In the Middle



Figure 1: Sinkhole, near Ein Gedi, Israel. Photograph by Adi Lavy



Figure 2: Mud bath, Mineral Beach, Photograph by Adi Lavy



Figure 3: Rocky Consumed by Potash, 3d recreated using photogrammetry. Photos by Ariel Shtern, Blimey Ltd.

Ages, however, attitudes towards the Dead Sea changed and it came to be considered satanic, actively hostile to anything with the breath of life in it.

As ironic as it sounds, the Dead Sea is dying; its waters are drying up, having severe consequences on the ecosystem, industry and wildlife in the area. As a result of years of human intervention and political neglect, the destruction is rapidly progressing. The water that used to feed it is diverted for industry, agriculture and domes-

tic use in both Israel, The Palestinian Territory and Jordan. The Geography of the land is changing very quickly, creating unique Geological phenomena. These changes usually take millions of years to develop but are now rapidly progressing to the amazement of scientists and researchers. The Destruction is so massive, so beautiful, so unreal, that it makes this place a wonderland for explorers, though unfortunately it is off limits to most. If things continue as they are, in less than 50 years, the Dead Sea as we know it will be gone for good.

It's an ecological crisis: the Dead Sea's water levels drop by 1.5 meters a year, depleted by an immutable barrage of water-politics. As its shores cave in, local economies and people's lives collapse with it. Paradoxically, its desiccation gives rise to hauntingly beautiful but ephemeral geological wonders, and reveals the eerie, enchanting world that used to be hidden beneath its waters.

The Dead Sea is the lowest land point on Earth. It is a hauntingly beautiful place, with striking vistas surrounded by desert, foothills, and mountains. Once Upon a Sea is a virtual reality experience allowing us to truly soak in the geological wonders - and disasters - of the region.

Once upon a sea is a contemporary portrait of the Dead Sea, documenting those who frequent it and it's massive destruction, while admiring the new beauty of the land revealed beneath the ruins. I would like to gift the Sea to people by granting access to these inaccessible beaches and creating a long lasting portrait of the Sea a minute before it's gone for good.

Each scene permits us to fully explore our characters' relationships with the environment, giving viewers control as to what they choose to focus on and interact with up close.

Photogrammetry was used to capture both the environments and artistic sculptures that make up the sceneries explored. Thousands of photos were carefully collected on location in dangerously disintegrating landforms. The photos were then processed in Reality Capture and 3D processed in 3DS and Maya before they were imported to Unity game engine. It was inside the game engine that the lighting design was added, and the sounds were spatially placed to create the full immersive experience. Inside the game engine, XR toolkit was used to enable users to interact with the assets created. Vis-à-vis 360 video documentation of the protagonists' journeys took place, and the videos were stitched and then edited and interwoven into the experience, to truly connect the users to the tragedy and demise of The Dead Sea.

Once Upon a Sea is now available for download on Steam and can also be viewed in a web browser for non VR users, with embedded 360 videos, at https://store.steampowered.com/app/1575480/Once_Upon_a_Sea/ and <http://deadseavr.com>.